

SQL Basics & Select

1. Display `employee_id`, `first_name`, `last_name`, and a new column `full_name` that is `first_name` and `last_name` concatenated with a comma between them (e.g., "John, Doe").

```
SELECT EMPLOYEE_ID, FIRST_NAME, LAST_NAME, FIRST_NAME || ', ' || LAST
_NAME AS FULL_NAME
FROM HR.EMPLOYEES
```

2. M2. List `employee_id`, `salary`, and a column `bonus_10_pct` showing 10% of salary (`salary * 0.10`)

```
SELECT EMPLOYEE_ID, SALARY, (SALARY+(SALARY*0.1)) AS bonus_10_p
ct
FROM HR.EMPLOYEES
```

3. M3. Show `employee_id`, `hire_date`, and a literal column `record_type` with value `'Employee'` for every row

```
SELECT EMPLOYEE_ID, HIRE_DATE, 'EMPLOYEE' AS RECORD_TYPE
FROM HR.EMPLOYEES
```

4. M4. For each employee, display `email` and a column `email_domain` set to the literal `'@company.com'`

```
SELECT EMAIL, '@COMPANY.COM' AS EMAIL_DOMAIN
FROM HR.EMPLOYEES;
```

5. M5. List `employee_id`, `salary`, `commission_pct`, and `effective_commission` where `effective_commission` is `NVL(commission_pct, 0)`

```
SELECT EMPLOYEE_ID, SALARY, COMMISSION_PCT, NVL ( COMMISSION_PCT,
0) AS EFFECTIVE_COMMISSION
FROM HR.EMPLOYEES
```

6. **M6.** Display `first_name` , `last_name` , and a column `initials` formed by the first character of `first_name` and the first character of `last_name` (e.g., "JD").

```
SELECT FIRST_NAME, LAST_NAME, SUBSTR ( FIRST_NAME, 1, 1) || SUBSTR (LA
ST_NAME, 1, 1) AS INITIALS
FROM HR.EMPLOYEES
```

7. **M7.** Show `employee_id` , `salary` , and `annual_salary` as `salary * 12`, and also `annual_plus_bonus` as `salary * 12 * 1.1` (10% bonus)

```
SELECT EMPLOYEE_ID, SALARY, SALARY*12 AS ANNUAL_SALARY, SALARY*1
2*1.1 AS BONUS
FROM HR.EMPLOYEES
```

8. **M8.** List all columns from `hr.departments` using explicit column names (`department_id`, `department_name`, `manager_id`, `location_id`)

```
SELECT DEPARTMENT_ID, DEPARTMENT_NAME, MANAGER_ID, LOCATION_ID
FROM HR.DEPARTMENTS
```

9. **M9.** From `hr.employees` , select `employee_id` and a column `description` that is the literal `'Emp#'` concatenated with `employee_id` (cast to string if needed: use `TO_CHAR(employee_id)`)

```
SELECT EMPLOYEE_ID, 'EMP#' || EMPLOYEE_ID AS DESCRIPTION
FROM HR.EMPLOYEES
```

10. **M10.** Display `job_id` , `salary` , and a column `salary_band` that is the literal `'Standard'` for every row

```
SELECT JOB_ID, SALARY, 'STANDARD' AS SALARY_BAND
FROM HR.EMPLOYEES
```

11. **M11.** List `employee_id`, `first_name`, `last_name`, and a column `display_name` as "Last, First" (`last_name`, comma space, `first_name`)

```
SELECT EMPLOYEE_ID, FIRST_NAME, LAST_NAME, LAST_NAME || ', ' || FIRST_NAME AS LAST_FIRST
FROM HR.EMPLOYEES
```

12. **M12.** Show `department_id` from `hr.departments` and a literal `1` as column `sort_order`

```
SELECT DEPARTMENT_ID, 1 AS SORT_ORDER
FROM HR.DEPARTMENTS
```

13. **M13.** From `hr.employees`, display `salary` and `monthly_net` as `salary * 0.85` (assuming 15% tax)

```
SELECT SALARY, SALARY*0.85 AS MONTHLY_NET
FROM HR.EMPLOYEES
```

- M14.** List `employee_id`, `commission_pct`, and `commission_display` where NULL `commission_pct` is shown as 0 using NVL

```
SELECT EMPLOYEE_ID, COMMISSION_PCT, NVL(TO_CHAR(COMMISSION_PCT), 'ABCG') AS COMMISSION_DISPLAY
FROM HR.EMPLOYEES
```

- M15.** Display `first_name`, `last_name`, `salary`, and a column `compensation` that is salary * (1 + NVL(`commission_pct`, 0))

```
SELECT FIRST_NAME, LAST_NAME, SALARY, SALARY*NVL(1+COMMISSION_PCT, 0) AS COMPANSATION
```

```
FROM HR.EMPLOYEES
```

M16. From `hr.departments` , list `department_name` and a literal column `region` with value `'HQ'` .

```
SELECT DEPARTMENT_NAME, 'HQ' AS REGION  
FROM HR.DEPARTMENTS
```

M17. Show `employee_id` , `hire_date` , and a column `years_label` with literal `'Years of service'`

```
SELECT EMPLOYEE_ID, HIRE_DATE, 'YEAR OF SERVICE' AS YEAR_LABEL  
FROM HR.EMPLOYEES
```

M18. List `employee_id` , `salary` , and `double_salary` as `salary * 2`.

```
SELECT EMPLOYEE_ID, SALARY, SALARY*2 AS DOUBLE_SALARY  
FROM HR.EMPLOYEES
```

M19. From `hr.employees` , display `manager_id` and a column `has_manager` that is the literal `'Yes'` when `manager_id` is not null and `'No'` when `manager_id` is null (use `NVL2: NVL2(manager_id, 'Yes', 'No')`).

```
SELECT MANAGER_ID, NVL2(MANAGER_ID, 'YES', 'NO') AS HAS_MANAGER  
FROM HR.EMPLOYEES
```

M20. Show `department_id` , `department_name` from `hr.departments` , and a calculated column `dept_code` as the first 3 characters of `department_name` (use `SUBSTR`).

```
SELECT DEPARTMENT_ID, DEPARTMENT_NAME, SUBSTR(DEPARTMENT_NAME,  
1, 3)  
FROM HR.DEPARTMENTS
```

HARD →

1. Display `employee_id`, `first_name`, `last_name`, `salary`, and a column `salary_rank_label` that is 'High' if salary ≥ 10000 , 'Medium' if salary ≥ 5000 and < 10000 , else 'Low'. Use CASE.

```
SELECT EMPLOYEE_ID, FIRST_NAME, LAST_NAME, SALARY, SALARY >= '10000' AS HIGH, SALARY >= '5000' AND SALARY < '10000' AS MEDIUM, SALARY < '5000' AS LOW
FROM HR.EMPLOYEES
```

2. List `employee_id`, `salary`, `commission_pct`, and `total_comp` as salary + (salary * NVL(commission_pct, 0)), formatted to 2 decimal places using ROUND(..., 2).

```
SELECT EMPLOYEE_ID, SALARY, COMMISSION_PCT, ROUND(SALARY * (1 + NVL(COMMISSION_PCT, 0)), 2) AS TOTAL_COMP
FROM HR.EMPLOYEES
```

3. From `hr.employees`, show `employee_id`, `email`, and `email_upper` as UPPER(email). Also show `email_length` as LENGTH(email)

```
SELECT EMPLOYEE_ID, EMAIL, UPPER(EMAIL) AS UPPER_EMAIL, LENGTH(EMAIL) AS EMAIL_LENGTH
FROM HR.EMPLOYEES
```

4. Display `department_id` from `hr.departments`, `department_name`, and a column `name_length` (number of characters in department_name).

```
SELECT DEPARTMENT_ID, LENGTH(DEPARTMENT_NAME) AS NAME_LENGTH
FROM HR.DEPARTMENTS
```

5. List `employee_id`, `first_name`, `last_name`, and a column `reverse_name` as last_name concatenated with first_name (no comma)

```
SELECT EMPLOYEE_ID, FIRST_NAME, LAST_NAME, LAST_NAME || FIRST_NAME AS REVERSE_NAME
```

```
FROM HR.EMPLOYEES
```

6. Show `employee_id`, `hire_date`, and a literal column `data_source` with value `'HR.EMPLOYEES'`.

```
SELECT EMPLOYEE_ID, HIRE_DATE, 'HR.EMPLOYEES' AS DATA_SOURCE  
FROM HR.EMPLOYEES
```

7. From `hr.employees`, display `job_id`, `salary`, and `salary_percentage` as $(\text{salary} / (\text{SELECT SUM(salary) FROM hr.employees})) * 100$, rounded to 2 decimals

```
SELECT JOB_ID, SALARY, ROUND(SALARY*100/(SELECT SUM(SALARY) FROM  
HR.EMPLOYEES), 2) AS SALARY_PERCENTAGE  
FROM HR.EMPLOYEES
```

8. H8. List `employee_id`, `first_name`, `last_name`, and a column `formal_name` as "Mr. " or "Ms. " (your choice) concatenated with `first_name` and `last_name`

```
SELECT EMPLOYEE_ID, FIRST_NAME, LAST_NAME, 'Mr.' || FIRST_NAME || LA  
ST_NAME AS FORMAL_NAME  
FROM HR.EMPLOYEES
```

9. Display `employee_id`, `salary`, and `annual_with_raise` as $(\text{salary} * 12) * 1.05$ (5% annual raise)

```
SELECT EMPLOYEE_ID, SALARY, SALARY*12*1.05 AS ANNUAL_RAISE  
FROM HR.EMPLOYEES
```

10. From `hr.departments`, show `department_id`, `department_name`, and a column `id_name` that concatenates `department_id` and `department_name` with a hyphen (e.g., "10-Administration"). Use `TO_CHAR(department_id)` for the number.

```
SELECT DEPARTMENT_ID, DEPARTMENT_NAME, DEPARTMENT_ID || '-' || DEPA  
RTMENT_NAME AS ID_NAME
```

```
FROM HR.DEPARTMENTS
```

11. List `employee_id`, `commission_pct`, and `commission_category`: 'Commissioned' if `commission_pct` is not null, 'Non-commissioned' if null. Use NVL2 or CASE.

```
SELECT EMPLOYEE_ID, COMMISSION_PCT, NVL2(COMMISSION_PCT, 'COMMISSIONED', 'NON-COMMISSION') AS  
COMMISSION_CATEGORY  
FROM HR.EMPLOYEES
```

12. H12. Show `employee_id`, `first_name`, `last_name`, `salary`, and a column `salary_expression` that is the literal string `'salary * 12'` (not the result of the calculation).

```
SELECT EMPLOYEE_ID, FIRST_NAME, LAST_NAME, SALARY, 'SAL*12' AS SAL  
ARY_EXPRESSION  
FROM HR.EMPLOYEES
```

13. From `hr.employees`, display `employee_id`, `job_id`, and a column `job_salary_label` as `job_id` concatenated with ':' and salary (e.g., "SA_REP:9000"). Use TO_CHAR(salary)

```
SELECT EMPLOYEE_ID, JOB_ID, JOB_ID || ':' || SALARY  
FROM HR.EMPLOYEES
```

14. List `employee_id`, `salary`, and `tax_bracket` as '20%' if `salary` >= 10000, '15%' if `salary` >= 5000, else '10%'. Use CASE.

```
SELECT EMPLOYEE_ID, SALARY,  
CASE  
  WHEN SALARY >= '10000' THEN '20%'  
  WHEN SALARY >= '5000' AND SALARY < '10000' THEN '15%'  
  ELSE '10%'  
END AS TAX_BRACKET  
FROM HR.EMPLOYEES
```

15. Display `department_id`, `department_name` from `hr.departments`, and a column `dept_info` that is "Department " followed by `department_id` and " - " and `department_name`.

```
SELECT DEPARTMENT_ID,DEPARTMENT_NAME,'DEPARTMENT' || DEPARTMENT_ID || ' - ' || DEPARTMENT_NAME AS
DEPT_NIF0
FROM HR.DEPARTMENTS
```

16. H16. From `hr.employees`, show `employee_id`, `first_name`, `last_name`, and `full_name_reversed` as `last_name`, space, `first_name` (e.g., "Doe John")

```
SELECT EMPLOYEE_ID,FIRST_NAME, LAST_NAME, LAST_NAME || ' ' || FIRST
_NAME AS NAME_REVERSED
FROM HR.EMPLOYEES
```

17. List `employee_id`, `salary`, `commission_pct`, and `effective_salary` as `salary` when `commission_pct` is null, otherwise `salary * (1 + commission_pct)`. Use `NVL2` or `CASE`.

```
SELECT EMPLOYEE_ID,SALARY,COMMISSION_PCT,NVL2(COMMISSION_PCT,SALARY*(1+COMMISSION_PCT),SA
LARY) AS EFFECTIVE_SALARY
FROM HR.EMPLOYEES
```

18. H18. Show `employee_id`, `hire_date`, and a column `hire_year` that is the year from `hire_date` (use `EXTRACT(YEAR FROM hire_date)` in Oracle).

```
SELECT EMPLOYEE_ID,HIRE_DATE,EXTRACT(YEAR FROM HIRE_DATE) AS Y
EAR
FROM HR.EMPLOYEES
```

19. From `hr.departments`, display `department_name` and a column `word_count` as the number of words (simplified: 1 + number of spaces, or use a formula based on `LENGTH` and `REPLACE` of spaces). For simplicity, use `LENGTH(department_name) - LENGTH(REPLACE(department_name, ' ', '')) + 1` if names have spaces.


```
SELECT DEPARTMENT_NAME, LENGTH(DEPARTMENT_NAME) - LENGTH(REPLACE
(DEPARTMENT_NAME, ' ', '')) + 1 AS COUNT
FROM HR.DEPARTMENTS
```

20. List `employee_id`, `first_name`, `last_name`, and a column `name_with_id` as `"[" || employee_id || "]" || first_name || " " || last_name` (e.g., "[100] Steven King").

```
SELECT EMPLOYEE_ID, FIRST_NAME, LAST_NAME, '[' || EMPLOYEE_ID || ']' ||
|| FIRST_NAME || ' ' || LAST_NAME NAME_WITHID
FROM HR.EMPLOYEES
```